Multimedia Datenbanken
aka Multimedia Databases, Multimedia Database

Prof. Dr. Harald Kosch, Prof. Dr. Marion Döller

1. Klausur im / exam in SS 25

Datum / Date: 08.08.25

Erlaubte Hilfsmittel / Permitted aids: Calculator

Dauer / Duration: 90 Minutes

Gesamtpunkte / Total points:

Achtung: Dies ist lediglich eine Niederschrift aus dem Gedächtnis, erstellt für die Fachschaft Informatik und Mathematik der Universität Passau. Es handelt sich nicht um die originale Klausur, es besteht keinerlei Anspruch auf Korrektheit oder Vollständigkeit!

1. Single-choice questions (8 pts)

2. Briefly state a well-known problem in training RNNs and name an associated mitigation (e.g., specific architectures or techniques) – min/max (?).

3. From a given list of steps (e.g., transform, quantization, entropy coding, etc.), identify the step that primarily reduces precision and explain why.

4. Given symbol probabilities (or frequencies), build the Huffman tree and derive the code for each symbol. Show the total expected code length.

5. Draw the provided plot/curve exactly: match the line styles and colors (e.g., purple line, black line, and other specified colors), label axes, and mark key points (from the lecture).

6. Given CMY values, compute the corresponding CMYK values.

7. RLE and zig-zag ordering

- (a) Apply Run-Length Encoding (RLE) to the provided coefficient sequence.

(b) Perform zig-zag scAn of an 8×8 block and write the resulting 1-D sequence.

8. Explain each concept in the context of image/video compression:

- Chroma subsampling
- Quantization
- Discrete Cosine Transform (DCT)

9. Carry out DCT computations (8x8) image in 3D then in 2D (similar to exercises)

10. (a) Define median-cut quantization.

(b) State one advantage of median-cut compared to the alternative method given in the prompt.

(c) Describe how the image/color space is recursively split in median-cut until the target palette size is reached.

11. List two important considerations for insertion and deletion in R-trees.

12. Perform the following on the provided R-tree and rectangles:

- Insert the given rectangles (show node choices and any splits).
- Delete the specified rectangles (handle underflow and reinsertion).
- Redraw the final tree and explain each step taken.

13. Write the SQL CREATE statements to define a type hierarchy with inheritance.

14. Explain what VARRAYs and nested TABLEs are (e.g., in Oracle or similar systems): storage characteristics, indexing implications, and typical use cases.